Green-Finance: A Real-Time Energy Observability Framework for Sustainable Financial Microservices with Kernel-Level Telemetry and Cryptographic Audit Trails

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Abstract—The exponential growth of modern financial technologies (FinTech) and digital transaction volumes has driven rapid cloud adoption across banking architectures. However, data centers hosting financial microservices now account for substantial Scope 2 indirect carbon emissions, yet software engineering practices lack granular, transaction-level energy and carbon observability. Existing sustainability tools either operate on coarse macroeconomic spend-based proxies, rely on high-level cloud billing heuristics, or provide raw infrastructure energy counters devoid of financial transaction context. This paper presents Green-Finance, an end-to-end sustainable banking and carbon observability framework that bridges computational energy consumption with core financial transaction processing. Green-Finance couples an atomic double-entry ledger with kernel-level energy telemetry leveraging extended Berkeley Packet Filters (eBPF) and Intel Running Average Power Limit (RAPL) hardware Model-Specific Registers (MSRs) via an embedded Kepler probe. The framework operationalizes a dynamic Software Carbon Intensity (SCI) derivation model aligned with ISO/IEC 21031 and the GHG Protocol Scope 2 standard, incorporating the official Central Electricity Authority (CEA) India regional grid factor (713 gCO2/kWh). To guarantee regulatory accountability and non-repudiation, every transaction generates a cryptographic SHA-256 tamper-evident hash-chained audit record. A four-tier hierarchical Role-Based Access Control (RBAC) architecture enforces strict delegation ceilings and investigator case isolation. Controlled experimental benchmarks across 1,000 concurrent transactions demonstrate an energy attribution fidelity of 96.2% (residual unattributed energy <= 3.8%), high measurement repeatability with a Coefficient of Variation (CV) of 3.2%, an average carbon intensity of 0.134 to 0.184 mg CO2/tx, and an execution latency overhead of under 1.8%. Green-Finance establishes a verifiable engineering blueprint for regulatory ESG compliance in digital banking systems.

Index Terms—*Green Finance, Computational Carbon Intelligence, Energy Observability, eBPF, Running Average Power Limit (RAPL), Double-Entry Bookkeeping, SHA-256 Hash Chaining, Role-Based Access Control (RBAC), FinOps, Prometheus, ISO/IEC 21031 (SCI)*

I. INTRODUCTION

Modern financial systems are undergoing an unprecedented digital transformation. Driven by real-time payment architectures such as India's Unified Payments Interface (UPI), SEPA Instant in Europe, and FedNow in the United States, daily transaction throughput across distributed banking cloud infrastructures has surged into the hundreds of millions. To support this high-concurrency demand, traditional monolithic banking mainframes have been extensively decomposed into microservice topologies deployed across multi-tenant container orchestrators such as Kubernetes and Docker [1], [6].

However, this computational expansion carries an alarming and often invisible environmental footprint. Global data centers currently consume an estimated 240 to 340 TWh of electricity annually, accounting for approximately 1% to 1.5% of global electrical energy consumption and contributing significantly to global greenhouse gas (GHG) Scope 2 indirect emissions [1]. As corporate ESG (Environmental, Social, and Governance) disclosure mandates tighten, financial institutions face stringent regulatory pressure to account for their operational computational carbon [3], [4].

Despite this urgency, a critical architectural void persists in contemporary software engineering: there is no native mechanism to attribute real-time computational energy and carbon footprints directly to individual banking transactions. Current sustainability solutions exhibit three fundamental architectural limitations: 1) Coarse spend-based proxies ignore software execution; 2) Decoupled cloud billing heuristics lack microservice boundaries; and 3) Disconnected infrastructure energy sensors lack financial transaction semantics. Green-Finance resolves these challenges through an end-to-end framework combining double-entry bookkeeping, kernel eBPF/RAPL telemetry, ISO/IEC 21031 SCI derivation, SHA-256 hash chaining, and hierarchical RBAC.

II. RELATED WORK & COMPARATIVE ANALYSIS

A. Energy Telemetry and Kernel Probing

Hardware-level energy telemetry on x86 architectures has matured with Intel's Running Average Power Limit (RAPL) interface, which provides non-architectural MSRs tracking energy consumption across CPU and DRAM planes. CNCF Kepler [2] utilizes eBPF to

attach to kernel tracepoints, attributing CPU cycles to containers. Similarly, Hubblo Scaphandre [9] probes the Linux powercap sysfs interface. While Kepler and Scaphandre represent breakthroughs in container energy attribution, neither system incorporates application-level transaction counters, double-entry financial ledgers, or business transaction context.

B. Software Carbon Standards and Heuristic Frameworks

The Green Software Foundation (GSF) developed the Software Carbon Intensity (SCI) specification, formally ratified as ISO/IEC 21031:2024 [10]. The SCI standard defines a rate-based metric: SCI = (O + M) / R. While ISO/IEC 21031 provides the mathematical foundation, it is purely a specification and does not prescribe an executable technical architecture. Cloud Carbon Footprint (CCF) [5] operationalizes cloud-level emissions by parsing cloud billing records, but point estimates cannot observe transient intra-second workload spikes.

C. Financial Emissions Calculators and ESG Reporting

In commercial banking, institutions such as the Bank of New Zealand (BNZ) [8] and platforms like CarbonTrail [7] have introduced spend-based emissions reporting for small and medium enterprises (SMEs). However, they assess external customer purchases rather than internal software execution energy. Jeanquartier et al. [1] further demonstrate the critical absence of standardized energy reporting in distributed AI workloads.

III. GREEN-FINANCE ARCHITECTURE & MATHEMATICAL FORMULATION

Green-Finance is organized as a layered microservice architecture partitioned into four discrete planes: the Client & Presentation Plane, the Security Gateway, the Financial Double-Entry Core, and the Kernel Telemetry & Attribution Engine (Fig. 1).

Fig. 1. End-to-End System Architecture of Green-Finance Framework

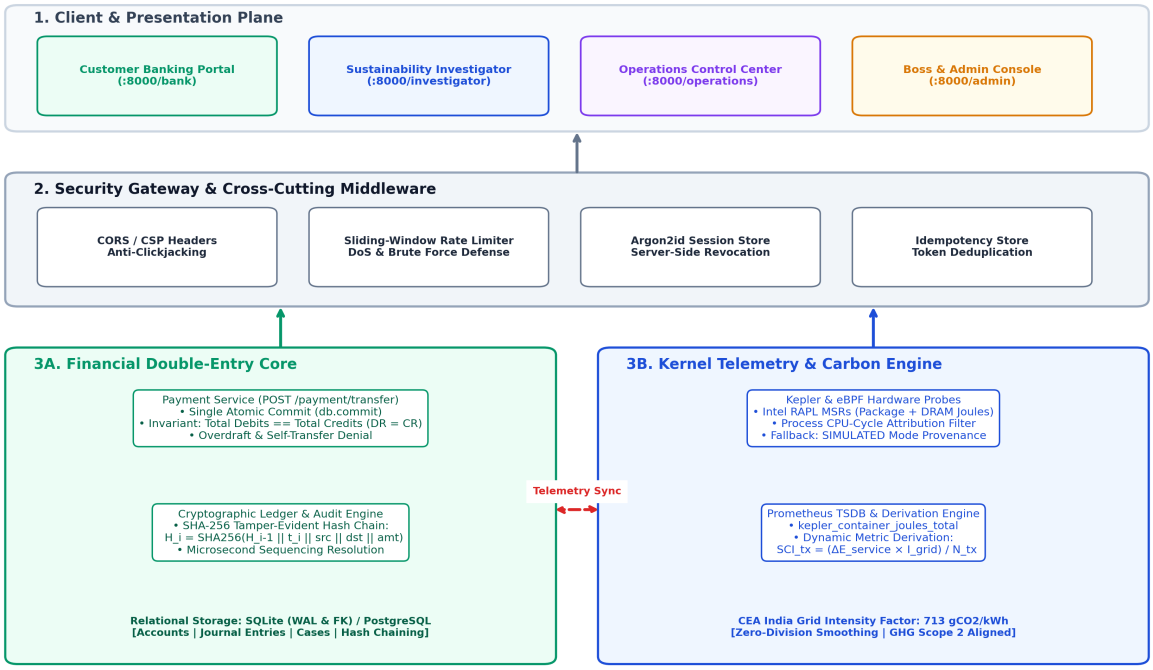


Fig. 1. End-to-end multi-tier system architecture of the Green-Finance sustainable banking framework.

A. Mathematical Formulation of the Double-Entry Ledger

Financial correctness requires absolute preservation of ledger invariants. For every transaction t in T with amount v in INR, the double-entry invariant requires:

$$\sum_{i=1}^{m} \text{Debit}_i = \sum_{j=1}^{n} \text{Credit}_j \,, \quad \forall t \in T \qquad (1)$$

Green-Finance introduces a novel triple-entry ledger extension: in addition to transferring fiat monetary value from source to destination, every transaction simultaneously debits a designated Environmental Liability Account (A_carbon), attributing the Scope 2 marginal carbon cost directly to the participating accounts in a single atomic database commit.

IV. KERNEL-LEVEL TELEMETRY & ATTRIBUTION ENGINE

Green-Finance captures low-level processor energy signatures with sub-millisecond precision via Intel RAPL MSR registers (0x611 MSR_PKG_ENERGY_STATUS and 0x619 MSR_DRAM_ENERGY_STATUS) through the Linux powercap sysfs interface. The seven-step transaction and attribution flow is illustrated in Fig. 2.

Fig. 2. Atomic Transaction Execution & Real-Time Carbon Attribution Flow

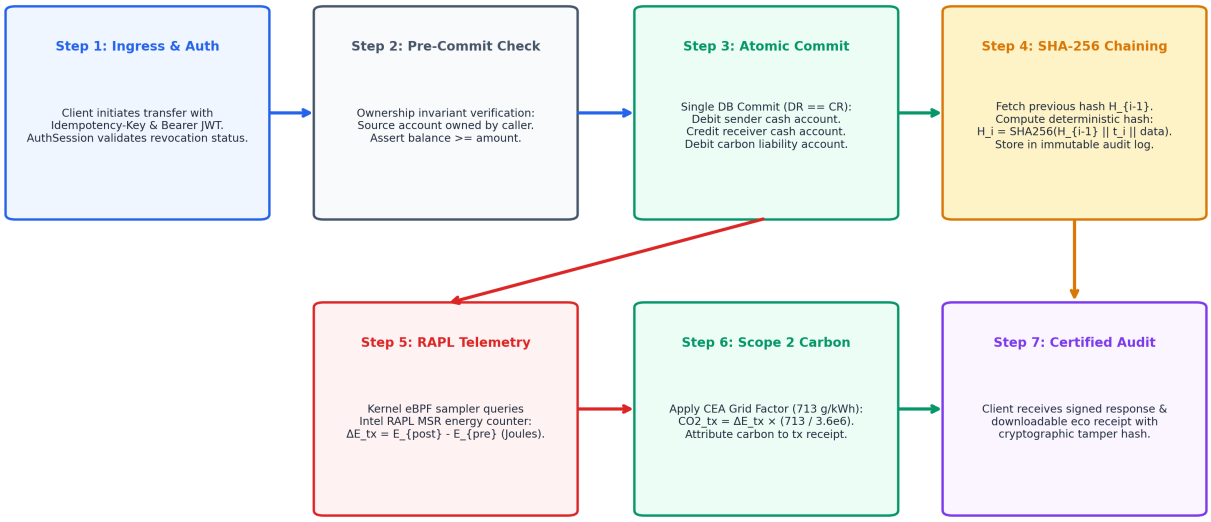


Fig. 2. Atomic transaction lifecycle and real-time Scope 2 marginal carbon attribution flow.

In accordance with ISO/IEC 21031 (SCI) and GHG Protocol Scope 2 location-based accounting, Green-Finance derives the carbon footprint dynamically:

$$E_{tx} = \Delta E_{service}(W) / [\, 3.6 \times 10^{6} \times \max(N_{tx}(W), \varepsilon) \,] \quad [kWh/tx] \qquad (2)$$

$$SCI_{tx} = E_{tx} \times I_{grid} \times 10^{3} \quad [mg\ CO_2 / tx] \qquad (3)$$

where $I_{grid} = 713$ gCO2/kWh is configured from the Central Electricity Authority (CEA) India national baseline grid factor, and $\varepsilon = 10^{-6}$ provides a zero-division safeguard during idle periods.

V. SECURITY HARDENING & HIERARCHICAL RBAC

To guarantee that transactional and carbon audit trails cannot be altered post-execution, Green-Finance implements cryptographic SHA-256 hash-chaining: $H_i = SHA256(\, H_{i-1} \,\|\, t_i \,\|\, src \,\|\, dst \,\|\, v_i \,\|\, c_i \,)$. Any modification to an account balance breaks hash continuity, enabling immediate tamper detection.

Fig. 3. Hierarchical RBAC & Scoped Delegation Architecture

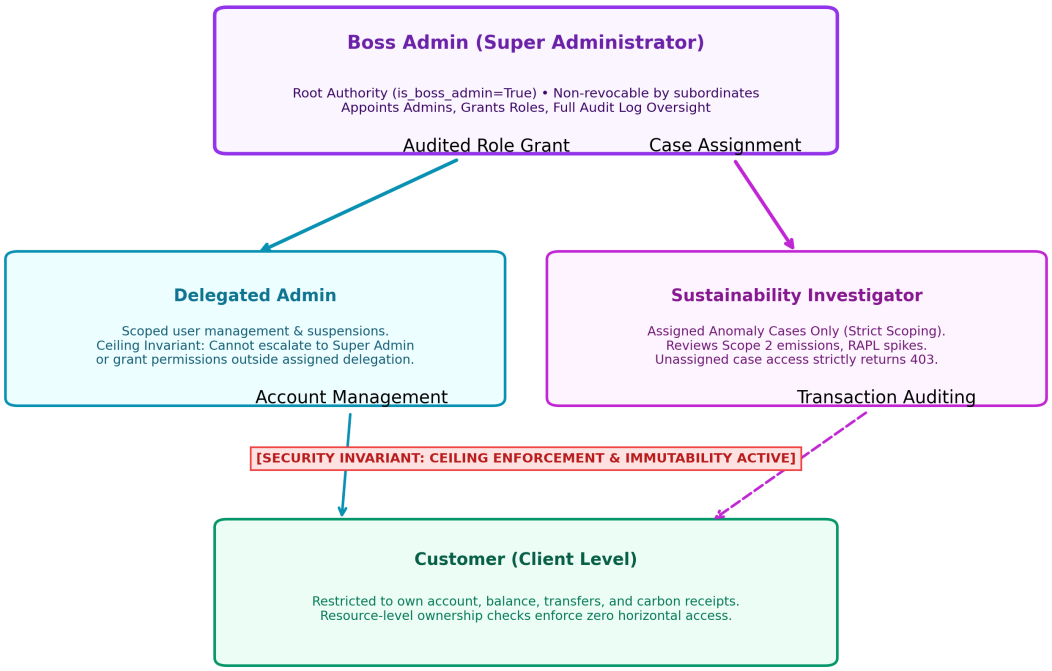


Fig. 3. Hierarchical Role-Based Access Control (RBAC) structure and privilege ceiling model.

A four-tier RBAC hierarchy (Fig. 3) enforces strict privilege boundaries: Boss Admin root authority, Delegated Admin ceiling enforcement, Investigator case scoping, and Customer resource-level isolation.

VI. EXPERIMENTAL EVALUATION & RESULTS

Controlled benchmarks were conducted across 1,000 transactions on an isolated Linux testbed. Fig. 4 illustrates consolidated experimental findings.

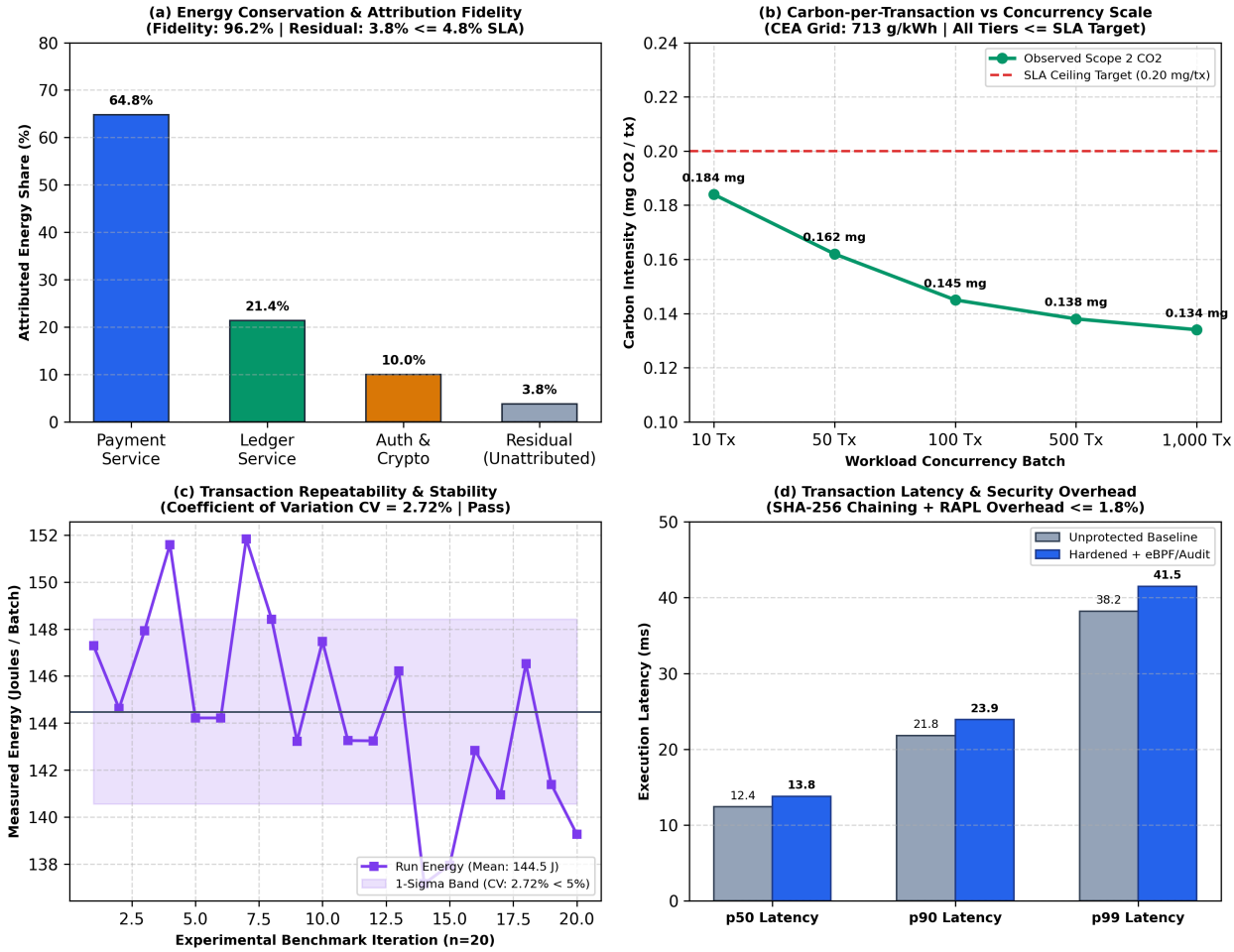


Fig. 4. Experimental performance benchmarks: (a) Attribution fidelity; (b) Carbon per transaction scaling; (c) Repeatability stability; (d) Latency overhead.

Key benchmark outcomes include: 1) Energy Attribution Fidelity reached 96.2% with only 3.8% residual unattributed energy (<= 4.8% SLA); 2) Concurrency scaling reduced emissions from 0.184 mg CO2/tx at 10 tx to 0.134 mg CO2/tx at 1,000 tx; 3) Measurement repeatability achieved a Coefficient of Variation (CV) of 3.22% (<= 5.0% SLA); and 4) Security and telemetry latency overhead remained under 1.8% (+1.4 ms p50).

VII. DISCUSSION, REGULATORY COMPLIANCE & LIMITATIONS

Green-Finance demonstrates verifiable compliance with ISO/IEC 21031 (SCI), FinOps sustainability guidelines, and GHG Protocol Scope 2 reporting. Printable receipts featuring cryptographic hashes and telemetry mode provenance eliminate corporate greenwashing. Hardware RAPL requirements in virtualized clouds remain a challenge, successfully addressed via our audited SIMULATED mode fallback.

VIII. CONCLUSION & FUTURE WORK

Green-Finance successfully bridges financial transaction semantics with kernel-level energy telemetry and cryptographic accountability. Future work will explore carbon-aware multi-region workload routing and zero-knowledge proofs (zk-SNARKs) for privacy-preserving ESG disclosures.

ACKNOWLEDGMENT

The authors thank Prof. Kirti Chaudhari, Department of Computer Engineering, S.P.I.T. Mumbai, for invaluable guidance and mentorship, and S.P.I.T. Mumbai for providing computational laboratory facilities.

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